EXPLORER

flask-ci-cd
- app.py
- requirements.txt
- Dockerfile
- .github / workflows
  - docker-deploy.yml

app.py

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def home():
    return "<h1>Hello from Flask -
            Deployed via GitHub Actions!</h1>"


if __name__ == "__main__":
    app.run(host="0.0.0.0', port=5000)

```

requirements.txt

```
Flask==2.3.3
```

Dockerfile

```
FROM python:3.10-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt
COPY app.py .
EXPOSE 5000
CMD ["python", "app.py"]
```

docker-deploy.yml

```yaml
name: Build and Deploy Flask App
on:
  push:
    branches: [ main ]
jobs:
  build-and-deploy:
    runs-on: ubuntu-latest
    steps:
      - name: Checkout code
        uses: actions/checkout@v4
      - name: Set up Docker Buildx
        uses: docker/setup-buildx-action@v3
      - name: Build Docker image
        run: docker build -t flask-app .
      - run: docker run -d -p 5000:5000 flask-app
```

username / flask-ci-cd

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Actions

All workflows

Build and Deploy Flask App

Build and Deploy Flask App

Build and Deploy Flask App #1

Re-run all jobs

Summary

Jobs

build-and-deploy

Run details

Usage

Workflow file

Triggered via push 2 minutes ago

username pushed 3f2d4a1 main

Status: Success

Total duration: 58s

Artifacts: –

build-and-deploy

succeeded 58s ago

Set up job	3s
Checkout code	2s
Set up Docker Buildx	6s
Build Docker image	15s
Run container	10s
Post job	2s

Terminal

```
root@controlplane:~/flask-ci-cd# docker ps
CONTAINER ID   IMAGE       COMMAND           CREATED          STATUS          PORTS                    NAMES
f3a2c7d4e9b8   flask-app   "python app.py"   12 seconds ago   Up 11 seconds   0.0.0.0:5000->5000/tcp   flask-app
root@controlplane:~/flask-ci-cd# curl http://localhost:5000/
<h1>Hello from Flask - Deployed via GitHub Actions!</h1>
root@controlplane:~/flask-ci-cd#
```

Flask App

http://localhost:5000

Hello from Flask - Deployed via GitHub Actions!